🧠 Prompt Engineering — Full Notes + MCQs (Simplified for Exam)

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This document is for revision purposes only — you must not depend on it as your only source for best exam results!

It is strongly recommended to **read the original GitHub repository / course materials thoroughly**. Use this guide to review, practice, and test yourself — not as the sole study resource.

https://github.com/panaversity/learn-low-code-agentic-ai/tree/main/00_prompt_engineering

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🌍 What Is Prompt Engineering?

Prompt engineering is the **art and technique of writing clear and purposeful instructions** for AI models like ChatGPT.
It's how we "talk" to AI in a way that gets us the best, most accurate, and useful responses.

It's both an **art (creative communication)** and a **science (structured experimentation)**.

Think of it like this:
🗣️ *You're the director*, the AI is your actor.
If your directions are vague — the "scene" falls apart.
If your directions are sharp — it performs perfectly.

💡 Why It Matters

1. **You don't need to code** — just clarity and logic.
2. **Good prompts improve performance** drastically.
3. It's an **iterative skill** — the more you test and tweak, the better you get.

4. It’s **becoming essential** for productivity in writing, research, coding, marketing, and more.

⚖️ Prompt Engineering vs Context Engineering

Aspect	Prompt Engineering	Context Engineering
Goal	Teach the model *how* to behave	Show the model *what* to know
Focus	Instructions, structure, tone	Knowledge, documents, tools, memory
Failure Mode	Vague instructions → poor answers	Missing info → hallucinations
Handled by	UX designers, app developers	Data teams, ML engineers
Example	“Summarize in 3 bullet points, be concise.”	“Attach last month’s sales data as reference.”

In short:
🧠 *Prompt = How to think*
📚 *Context = What to know*

You almost always need both.
Good prompts without good context = wrong info.
Good context without clear prompts = messy outputs.

🧩 How LLMs Work (in simple words)

Large Language Models (LLMs) like GPT are basically **super-advanced autocompleters.**

They:

1. Take your text input (prompt).
2. Predict the next most likely word (token).
3. Keep predicting until the response is done.
4. Base it all on patterns they learned from huge text datasets.

They don't "understand" like humans — they *simulate understanding* by predicting what sounds right based on training data.

That's why your **prompt quality = response quality**.

⚙️ Model Settings (Configuration Essentials)

🔹 Temperature (0–1)

Controls *creativity vs accuracy*.

Range	Behavior	Example Use
0–0.3	Deterministic, focused, logical	Math, data analysis, factual Qs
0.4–0.7	Balanced creativity	Essays, summaries, explanations
0.8–1.0	Creative, unpredictable	Poems, brainstorming, storytelling

🧠 *Rule:*
→ Lower = "safe and strict."
→ Higher = "creative and free."

🔹 Top-K and Top-P (Nucleus Sampling)

Control *how random* the AI's word choice can be.

- **Top-K:** AI picks from top K most likely words.
- **Top-P:** AI picks from a group of words that together make up P% of probability.

👉 Example:
If Top-K = 10, it only chooses from top 10 likely next words.
If Top-P = 0.9, it only considers words that make up 90% of the probability mass.

🔹 Token Limit

- Controls the *maximum output length*.
- Long answers = more tokens = more cost.

- Always set appropriately for your use case.

💬 Fundamental Prompting Techniques

1️⃣ Zero-Shot Prompting

No examples, just a direct question.

🧠 *Example:*
"Translate this sentence into Urdu: 'I love learning.'"

When to use:

- Tasks are simple.
- AI already knows the topic.
- You want quick, one-off answers.

2️⃣ One-Shot Prompting

Give *one example* before the real task.

🧠 *Example:*
English: "Hello."
French: "Bonjour."
English: "Good night."
French: ?

When to use:
When you want AI to follow a certain *format* or *style.*

3️⃣ Few-Shot Prompting

Give *multiple examples* (3–5 usually).

🧠 *Example:*
Feedback → JSON structured data.
Helps AI learn the pattern properly.

When to use:

- Classification or structured output.
- When consistency and formatting matter.

4️⃣ System Prompting

Set AI’s personality or role permanently for a task.

🧠 *Example:*
“You are a helpful HR assistant who writes short, polite replies to employees.”

5️⃣ Role Prompting

Ask AI to “act as” an expert.

🧠 *Example:*
“Act as a nutritionist and suggest a healthy breakfast plan.”

6️⃣ Contextual Prompting

Feed it relevant *background info*.

🧠 *Example:*
Context: “Target audience is beginners who don’t code.”
Task: “Explain what an API is.”

🚀 Advanced Prompting Strategies

🧩 Chain of Thought (CoT)

Tell AI to reason step-by-step.
Best for: logic, problem solving, math, reasoning.

🧠 Example:
“Let’s solve this step by step. If I was 6 and my sister was half my age…”

✅ Makes reasoning explicit and reduces mistakes.

🔁 Self-Consistency

Ask multiple times → choose the most common answer.
Helps reduce random reasoning errors.

🧠 Example:
Run same math problem 3 times; pick the answer that repeats.

🧭 Step-Back Prompting

Ask general → then specific.
This gives the AI context before details.

🧠 Example:

1. What are the main causes of slow websites?

2. Now, fix mine using those causes.

⚙️ ReAct (Reasoning + Acting)

AI alternates between *thinking* and *doing.*
It plans, searches, observes, and then answers.

🧠 Example:
Thought → "Need Tokyo population."
Action → Search
Observation → "Tokyo = 37.4M"
Final → "Tokyo has 13.8M more than NYC."

🌳 Tree of Thoughts (ToT)

AI creates *multiple ideas or plans* → evaluates → merges best parts.

🧠 Example:
3 marketing strategies → compare pros and cons → choose best.

📈 A/B Testing in Prompt Engineering (Expanded)

A/B testing means creating **two or more versions** of the same prompt to compare which performs better.

You change *only one thing at a time* — like wording, example order, or output format — and measure the results.

🧠 Example:
Prompt A: "Summarize this article in 3 bullet points."
Prompt B: "Give a short summary of the article using bullet points and one concluding sentence."

Then you compare:

- Which one followed the structure better?
- Which one gave clearer results?

This helps you **scientifically improve prompts**, instead of guessing.

✅ Best Practices:

- Test on same data.
- Log results in a table.
- Track accuracy, clarity, tone, completeness.

⚠️ Common Pitfalls

Problem	What Happens	Fix
Vague prompt (ambiguous)	Random, messy output	Be specific & give structure
Conflicting instructions (contradict)	Model gets confused	Keep tone and goals consistent
Too many don'ts(constraints)	AI freezes	Focus on what you *do* want
Too many tokens	Output cut off	Limit or summarize context
No testing	Weak results	A/B test variations

🧩 Practical Example: Data Analysis

Prompt:
"Analyze customer feedback and tell me:

1. % of positive/negative comments
2. 3 main issues
3. Suggestions for improvement
   Return answer as a report."

✅ This shows structure, tone, and clear expected output.

🧾 THEORY MCQs (with answers)

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1. What is the main goal of prompt engineering?
   A) To design data pipelines for AI
   B) To write perfect AI code
   C) To craft effective instructions for AI behavior
   D) To increase model size
   Correct Answer: C

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2. Context engineering focuses on:
   A) Choosing AI temperature
   B) Providing relevant facts or documents
   C) Improving grammar
   D) Reducing output tokens
   Correct Answer: B

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3. What does a low temperature (e.g., 0.1) do?
   A) Makes output random
   B) Makes responses creative
   C) Makes responses consistent and factual
   D) Makes AI silent

Correct Answer: C

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4. Few-shot prompting means:
   A) Asking many questions at once
   B) Giving several examples before asking the main task
   C) Using no examples
   D) Using tools for reasoning
   Correct Answer: B

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5. Which of the following best describes Chain of Thought prompting?
   A) Asking general then specific questions
   B) Encouraging step-by-step reasoning
   C) Using multiple prompts at once
   D) Generating random ideas
   Correct Answer: B

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6. Step-Back prompting is useful when:
   A) You want the AI to generate art
   B) You need context before a specific solution
   C) You want faster answers
   D) You're testing token length
   Correct Answer: B

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7. ReAct prompting involves:
   A) Reacting emotionally
   B) Reasoning + taking actions like searches or calculations
   C) Random text generation
   D) Repeating prompts
   Correct Answer: B

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8. Tree of Thoughts is best for:
   A) Simple factual questions
   B) Multi-choice quizzes
   C) Creative or strategic decision making

D) Token optimization
Correct Answer: C

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9. Which statement is true about A/B testing?
   A) It's about measuring accuracy of models
   B) It's about comparing two prompt variations
   C) It's about measuring token usage
   D) It's about increasing randomness
   Correct Answer: B

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10. In self-consistency, the AI:
   A) Gives random answers
   B) Produces multiple answers and we pick the most common one
   C) Writes faster
   D) Uses shorter tokens
   Correct Answer: B

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11. System prompting helps by:
   A) Defining AI's role and behavior style
   B) Reducing token length
   C) Generating multiple answers
   D) Creating hallucinations
   Correct Answer: A

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12. "Attach company policy and answer only from it." is an example of:
   A) Zero-shot prompting
   B) Contextual prompting
   C) Step-back prompting
   D) Role prompting
   Correct Answer: B

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13. The biggest failure mode of bad context is:
   A) Overly detailed instructions
   B) Hallucination or outdated answers

C) Creative overflow
D) System crash
Correct Answer: B

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14. When to use temperature = 0.9?
A) For factual data
B) For creative writing
C) For math
D) For summarization
Correct Answer: B

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15. Token limits affect:
A) Grammar correctness
B) Output length and cost
C) Model training
D) Memory usage only
Correct Answer: B

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16. A “prompt contract” defines:
A) Output format and behavior expectations
B) Cost of model usage
C) Data policy
D) None of these
Correct Answer: A

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17. In prompt chaining:
A) All tasks are done in one step
B) Tasks are divided into smaller sub-prompts
C) The AI repeats the same answer
D) Tokens are merged
Correct Answer: B

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18. Which of these is a *common pitfall*?
A) Clear structure

B) Specific tone
C) Contradictory instructions
D) Defined output format
Correct Answer: C

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19. In context engineering, “RAG” stands for:
A) Random AI Generator
B) Retrieval-Augmented Generation
C) Recurrent Action Grouping
D) Read And Generate
Correct Answer: B

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20. Prompt engineering is mainly about:
A) What to show the AI
B) How to ask the AI
C) How to code the AI
D) When to stop the AI
Correct Answer: B

🎬 SCENARIO-BASED MCQs

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1. You write: “Summarize this article in 2 lines.”
AI writes 10 lines. What likely went wrong?
A) Temperature too high
B) Prompt lacked clear constraint enforcement
C) Too much context
D) Context missing
Correct Answer: B

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2. You add “Explain like I’m five” to a technical prompt. You just used:
A) Step-back prompting
B) Role prompting (teacher)
C) Contextual prompting
D) Self-consistency

Correct Answer: B

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3. You gave 3 examples before asking for a final response. That's:
   A) One-shot prompting
   B) Few-shot prompting
   C) Step-back prompting
   D) Chain of Thought
   Correct Answer: B

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4. You test two prompts: one with "summarize in bullets," another with "summarize in paragraph." You compare which works better. You're doing:
   A) Context testing
   B) A/B testing
   C) Token testing
   D) Random sampling
   Correct Answer: B

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5. You tell the AI: "Think aloud before answering." This activates:
   A) Step-back prompting
   B) Chain of Thought reasoning
   C) Contextual prompting
   D) Role prompting
   Correct Answer: B

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6. Your AI keeps using wrong info because it wasn't given recent data. Problem?
   A) Prompt structure
   B) Missing context (RAG failure)
   C) Too low temperature
   D) Wrong tone
   Correct Answer: B

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7. You want creative story writing — which setup works best?
   A) Temperature 0.1
   B) Temperature 0.9

C) Step-back prompt
D) Context retrieval
Correct Answer: B

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8. You ask: “What’s your plan to answer?” before “Now answer it.” You used:
   A) Chain of Thought
   B) Step-Back prompting
   C) Few-shot
   D) Tree of Thoughts
   Correct Answer: B

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9. Your model explores 3 marketing ideas and then combines them. This is:
   A) Tree of Thoughts
   B) Self-consistency
   C) ReAct
   D) Zero-shot
   Correct Answer: A

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10. The AI reasons, searches online, and then answers with evidence. This is:
    A) Role prompting
    B) ReAct prompting
    C) Contextual prompting
    D) Few-shot prompting

Thanks for reading best of luck!!!